

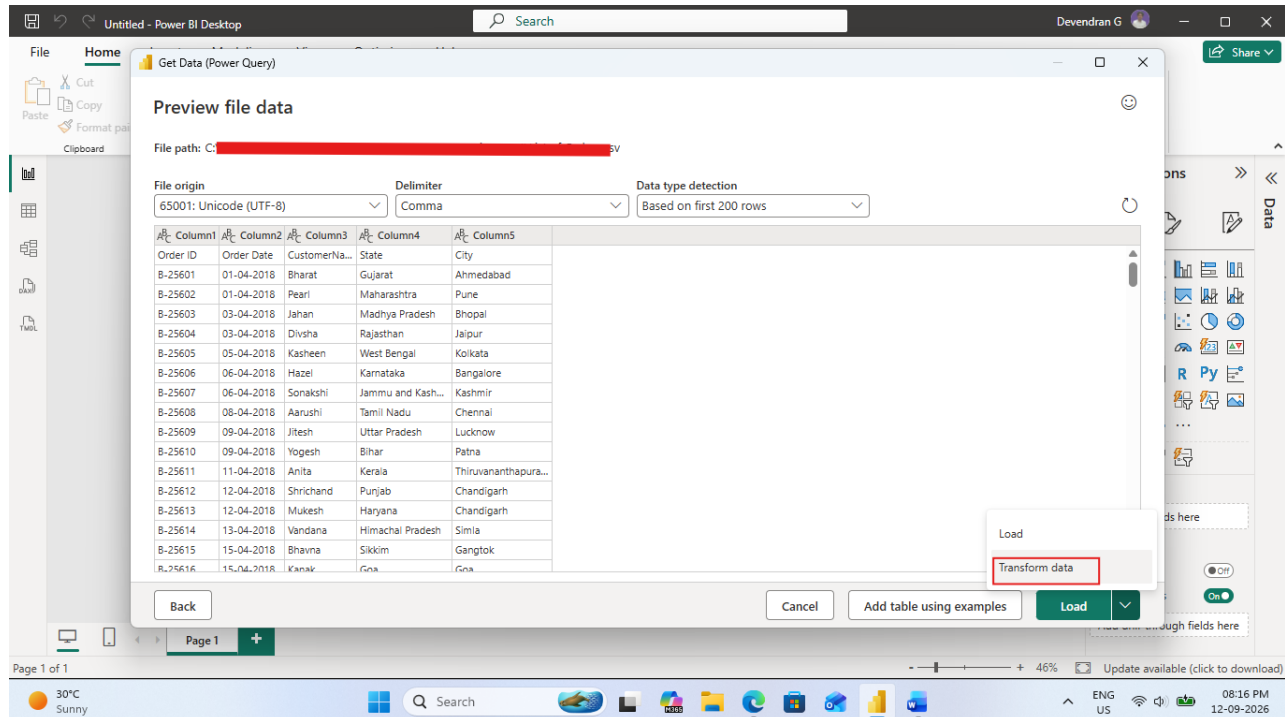
POWER BI ASSIGNMENT 1

Data Transformation & Data Modeling

Evidence document — questions, relevant screenshots and short explanations

Question / Requirement

Import “List of Orders.csv” into Power BI, open “List of Orders” in Power Query Editor, and import “Order Details.csv” and “Sales target.csv” into Power Query Editor.



Explanation: The required CSV files were imported into Power Query for transformation and analysis.

Question / Requirement

Restrict the “List of Orders” table to only the first 500 rows.

The screenshot shows the Power Query Editor interface. The main query is 'List of Orders' with the formula bar containing: `= Table.TransformColumnTypes(#"Promoted Headers",{{"Order ID", type text}, {"Order Date", type date}},`. A dialog box titled 'Keep Top Rows' is open, asking 'Specify how many rows to keep.' with a text box containing '500'. The background table shows columns: Order ID, Order Date, CustomerName, State, and City. The status bar indicates '5 COLUMNS, 560 ROWS'.

The screenshot shows the Power Query Editor after applying the 'Keep Top Rows' step. The formula bar now contains: `= Table.FirstN(#"Changed Type", 500)`. A tooltip for the 'Order ID' column shows '500 (100%) Valid'. The 'APPLIED STEPS' pane on the right shows the sequence: Source, Promoted Headers, Changed Type, and Kept First Rows. The status bar indicates '5 COLUMNS, 500 ROWS'.

Explanation: The List of Orders query was limited to the first 500 rows using Keep Top Rows. The final query shows 500 rows.

Question / Requirement

Ensure the “Order Date” column in the “List of Orders” table is set to data type 'Date'.

The screenshot shows the Power Query Editor interface. The ribbon at the top includes tabs for File, Home, Transform, Add Column, View, Tools, and Help. The 'Transform' tab is active, and the 'Data Type: Date' dropdown is highlighted in the ribbon. The main area displays a table with columns: Order ID, Order Date, CustomerName, State, and City. The 'Order Date' column is highlighted with a red box. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list, which includes 'Changed Type' and 'Kept First Rows'. The 'Properties' pane shows the 'Name' as 'List of Orders'.

Order ID	Order Date	CustomerName	State	City
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
B-25602	01-04-2018	Pearl	Maharashtra	Pune
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
B-25606	06-04-2018	Hazel	Karnataka	Bangalore
B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
B-25610	09-04-2018	Yogesh	Bihar	Patna
B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
B-25616	15-04-2018	Kanak	Goa	Goa
B-25617	17-04-2018	Sagar	Nagaland	Kohima
B-25618	18-04-2018	Maniu	Andhra Pradesh	Hyderabad

Explanation: Order Date was changed to the Date data type in Power Query.

Question / Requirement

Change the data type of “Amount” and “Target” columns to ‘Fixed Decimal Number’.

The screenshot shows the Power Query Editor interface. The main data table has columns: Order ID, Amount, Profit, Quantity, and Category. The 'Amount' column is selected, and the 'Data Type' dropdown menu is open, showing 'Fixed decimal number' as the selected option. The formula bar shows the transformation: `= Table.TransformColumnTypes(#"Promoted headers", {{{"Order ID", type text}, {"Amount", Int64.Type}, {"Profit", Int64.Type}, {"Quantity", Int64.Type}, {"Category", type text}}})`. The right-hand pane shows the 'APPLIED STEPS' list with 'Changed column type' at the bottom.

Order ID	Amount	Profit	Quantity	Category
B-25601	1275	-1148	7	Furniture
B-25601	66	-12	5	Clothing
B-25601	8	-2	3	Clothing
B-25601	80	-56	4	Electronics
B-25602	168	-111	2	Electronics
B-25602	424	-272	5	Electronics
B-25602	2617	1151	4	Electronics
B-25602	561	212	3	Clothing
B-25602	119	-5	8	Clothing
B-25603	1355	-60	5	Clothing
B-25603	24	-90	1	Furniture
B-25603	193	-166	3	Clothing
B-25603	180	5	3	Clothing
B-25603	116	16	4	Clothing
B-25603	107	36	6	Clothing
B-25603	12	1	2	Clothing
B-25603	38	18	1	Clothing

The screenshot shows the Power Query Editor interface. The main data table has columns: Month of Order Date, Category, and Target. The 'Target' column is selected, and the 'Data Type' dropdown menu is open, showing 'Fixed decimal number' as the selected option. The formula bar shows the transformation: `= Table.TransformColumnTypes(#"Changed column type", {{{"Target", Currency.Type}}})`. The right-hand pane shows the 'APPLIED STEPS' list with 'Changed Type' at the bottom.

Month of Order Date	Category	Target
18-04-2026	Furniture	10,400.00
18-05-2026	Furniture	10,500.00
18-06-2026	Furniture	10,600.00
18-07-2026	Furniture	10,800.00
18-08-2026	Furniture	10,900.00
18-09-2026	Furniture	11,000.00
18-10-2026	Furniture	11,100.00
18-11-2026	Furniture	11,300.00
18-12-2026	Furniture	11,400.00
19-01-2026	Furniture	11,500.00
19-02-2026	Furniture	11,600.00
19-03-2026	Furniture	11,800.00
18-04-2026	Clothing	12,000.00
18-05-2026	Clothing	12,000.00
18-06-2026	Clothing	12,000.00
18-07-2026	Clothing	14,000.00
18-08-2026	Clothing	14,000.00
18-09-2026	Clothing	14,000.00

Explanation: Amount and Target were converted to the Fixed Decimal Number/Currency representation required for the assignment.

Question / Requirement

Format the “CustomerName” column into proper case, ensuring consistent capitalization for each word.

The screenshot shows the Power Query Editor interface. The main data table has columns: Order ID, Order Date, CustomerName, State, and City. The 'CustomerName' column is selected, and a right-click context menu is open. The 'Transform' option is chosen, leading to a submenu where 'Capitalize Each Word' is selected. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list, which includes 'Source', 'Promoted Headers', 'Changed Type', and 'Kept First Rows'. The status bar at the bottom indicates '5 COLUMNS, 500 ROWS' and 'Column profiling based on entire data set'.

Order ID	Order Date	CustomerName	State	City
1	01-04-2018	Bharat	medabad	
2	01-04-2018	Pearl	ne	
3	03-04-2018	Jahan	opai	
4	03-04-2018	Divsha	pur	
5	05-04-2018	Kasheen	ikata	
6	06-04-2018	Hazel	ngalore	
7	06-04-2018	Sonakshi	shmir	
8	08-04-2018	Aarushi	ennai	
9	09-04-2018	Jitesh	know	
10	09-04-2018	Yogesh		
11	11-04-2018	Anita		
12	12-04-2018	Shrichand		
13	12-04-2018	Mukesh		
14	13-04-2018	Vandana		
15	15-04-2018	Bhavna		
16	15-04-2018	Kanak		
17	17-04-2018	Sagar		
18	18-04-2018	Manju		
19	18-04-2018	Ramesh		
20	20-04-2018	Sarita		
21	20-04-2018	Deepak		
22	22-04-2018	Monisha		
23	22-04-2018	Atharv		

Explanation: CustomerName was transformed using Capitalize Each Word (Proper Case) to standardize capitalization.

Question / Requirement

Merge the “State” and “City” columns to create a new column named “Location” in the format ‘City, State’.

The screenshot shows the Power Query Editor interface. The main table has the following data:

Order ID	Order Date	CustomerName	State	City
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
B-25602	01-04-2018	Pearl	Maharashtra	Pune
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
B-25606	06-04-2018	Hazel	Karnataka	Bangalore

The 'Merge Columns' dialog box is open, showing the 'Separator' dropdown set to '--Custom--' and the 'New column name (optional)' field containing 'Location'. The 'OK' button is highlighted.

Explanation: The State and City columns were merged into a new Location column. Important: the current screenshot shows the values in State, City order; the assignment specifically requires City, State. This order should be corrected in the PBIX before final submission.

Question / Requirement

Create a new custom column named “Profit Margin” as the percentage of “Profit” divided by “Amount”.

Custom Column

Add a column that is computed from the other columns.

New column name
Profit Margin

Custom column formula
=[Profit]/[Amount]

Available columns
Order ID
Amount
Profit
Quantity
Category
Sub-Category

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK Cancel

Change Type

Percentage

Explanation: A custom Profit Margin column was created using Profit divided by Amount and then formatted as a percentage.

Question / Requirement

Add a new conditional column named “Profit Status”: Profit < 0 = “Loss”, Profit = 0 = “Break-Even”, and Profit > 0 = “Profit”.

The screenshot shows the Power Query Editor interface. The 'Add Conditional Column' dialog box is open, allowing the user to define a new column based on existing data. The dialog is configured as follows:

- New column name:** Profit Status
- Column Name:** Profit
- Operator:** is less than
- Value:** 0
- Output:** Loss
- Else If:** Profit
- Operator:** equals
- Value:** 0
- Output:** Break-Even
- Else:** Profit

The background data table is as follows:

	Profit	Quantity	Category	Sub-Category	% Profit Margin
1	1,275.00	-1148	7 Furniture	Bookcases	-90.04%
2	66.00	-12	5 Clothing	Stole	-18.18%
3	8.00	-2	3 Clothing	Hankchief	-25.00%

Explanation: Profit Status was created with the required three conditions: Loss, Break-Even and Profit.

Question / Requirement

Merge the “List of Orders” and “Order Details” tables into a new single table named “Orders Data” based on the “Order ID” relationship.

Power Query Editor - Merge dialog box

Select tables and matching columns to create a merged table.

Table 1: List of Orders

Order ID	Order Date	CustomerName	Location
B-25601	01-04-2018	Bharat	Gujarat, Ahmedabad
B-25602	01-04-2018	Pearl	Maharashtra, Pune
B-25603	03-04-2018	Jahan	Madhya Pradesh, Bhopal
B-25604	03-04-2018	Divsha	Rajasthan, Jaipur
B-25605	05-04-2018	Kasheen	West Bengal, Kolkata

Table 2: Order Details

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status
B-25601	1,275.00	-1148	7	Furniture	Bookcases	-90.04%	Loss
B-25601	66.00	-12	5	Clothing	Stole	-18.18%	Loss
B-25601	8.00	-2	3	Clothing	Hankerchief	-25.00%	Loss
B-25601	80.00	-56	4	Electronics	Electronic Games	-70.00%	Loss
B-25602	168.00	-111	2	Electronics	Phones	-66.07%	Loss

Join Kind: Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

Fuzzy matching options

✓ The selection matches 500 of 500 rows from the first table.

OK Cancel

Power Query Editor - Expand dialog box

Table.NestedJoin("#List of Orders", {"Order ID"}, {"Order Details", {"Order ID"}, "Order Details",

Search Columns to Expand

Expand Aggregate

(Select All Columns)

☒ Order ID

☒ Amount

☒ Profit

☒ Quantity

☒ Category

☒ Sub-Category

☒ Profit Margin

☒ Profit Status

☐ Use original column name as prefix

OK Cancel

Explanation: List of Orders was merged with Order Details using Order ID with a Left Outer join, and the required Order Details fields were expanded into Orders Data.

Question / Requirement

Identify missing values in the data and determine a strategy to address them.

Explanation: Missing values were checked during data cleaning. No missing values requiring imputation or replacement were identified, so no imputation was applied.

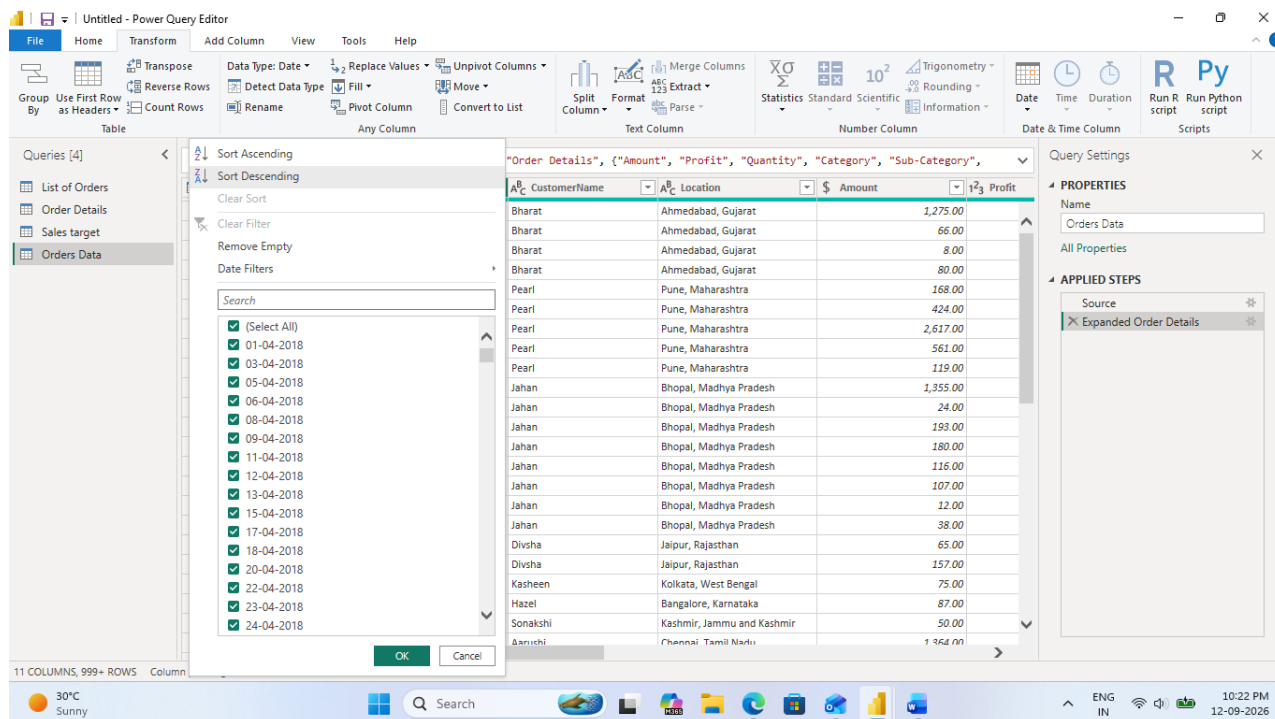
Question / Requirement

Check for duplicate rows and define a strategy to handle duplicates.

Explanation: Duplicate rows were checked. No exact duplicate rows requiring removal were identified. Repeated Order IDs are valid because one order can have multiple detail records.

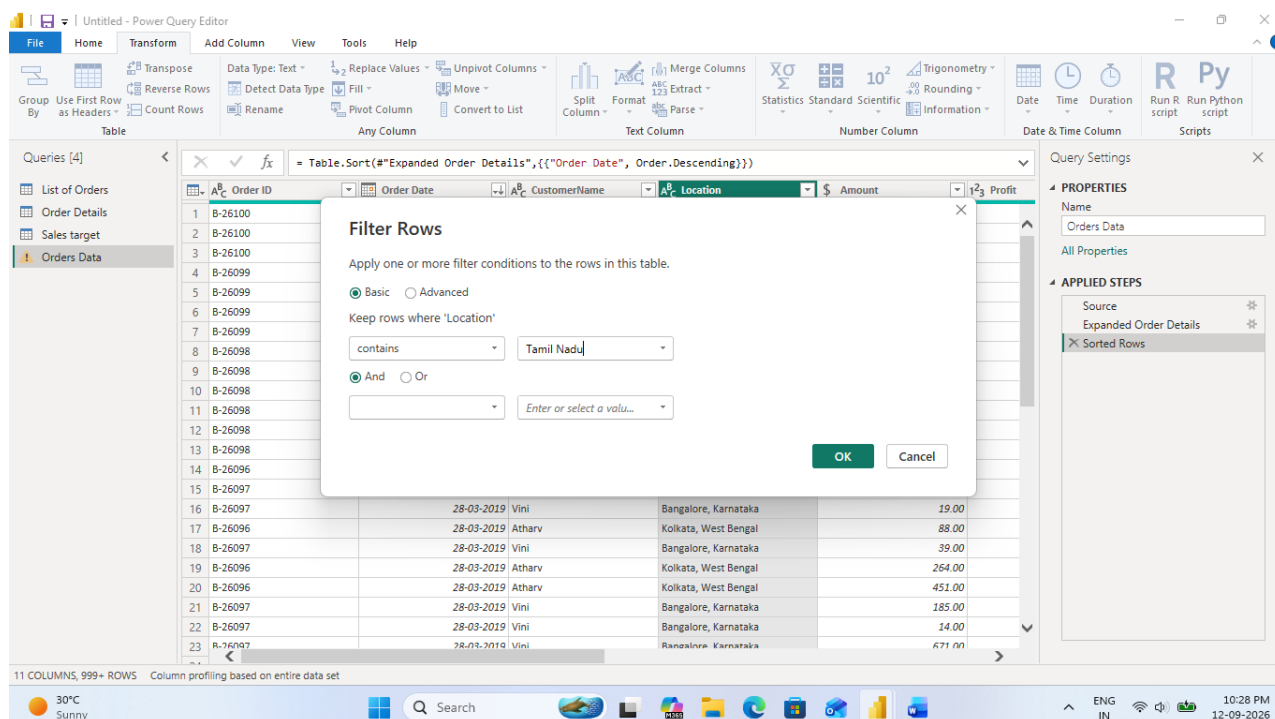
Question / Requirement

In the “Orders Data” table, sort the orders by Order Date in descending order and filter the orders to focus only on a specific state, such as Tamil Nadu.



Power Query Editor interface showing the initial state of the 'Orders Data' table. The table is sorted by 'Order Date' in descending order. A date filter is applied, showing a list of dates from 01-04-2018 to 24-04-2018. The table has 11 columns and 999 rows.

CustomerName	Location	Amount	Profit
Bharat	Ahmedabad, Gujarat	1,275.00	
Bharat	Ahmedabad, Gujarat	66.00	
Bharat	Ahmedabad, Gujarat	8.00	
Bharat	Ahmedabad, Gujarat	80.00	
Pearl	Pune, Maharashtra	168.00	
Pearl	Pune, Maharashtra	424.00	
Pearl	Pune, Maharashtra	2,617.00	
Pearl	Pune, Maharashtra	561.00	
Pearl	Pune, Maharashtra	119.00	
Jahan	Bhopal, Madhya Pradesh	1,355.00	
Jahan	Bhopal, Madhya Pradesh	24.00	
Jahan	Bhopal, Madhya Pradesh	193.00	
Jahan	Bhopal, Madhya Pradesh	180.00	
Jahan	Bhopal, Madhya Pradesh	116.00	
Jahan	Bhopal, Madhya Pradesh	107.00	
Jahan	Bhopal, Madhya Pradesh	12.00	
Jahan	Bhopal, Madhya Pradesh	38.00	
Divsha	Jaipur, Rajasthan	65.00	
Divsha	Jaipur, Rajasthan	157.00	
Kasheen	Kolkata, West Bengal	75.00	
Hazel	Bangalore, Karnataka	87.00	
Sonakshi	Kashmir, Jammu and Kashmir	50.00	
Aashish	Chennai, Tamil Nadu	1,564.00	



Power Query Editor interface showing the 'Orders Data' table after sorting and filtering. The table is sorted by 'Order Date' in descending order. A filter is applied to the 'Location' column, showing only rows where the location is 'Tamil Nadu'. The table has 11 columns and 999 rows.

Filter Rows dialog box:

Apply one or more filter conditions to the rows in this table.

Basic (selected) Advanced

Keep rows where 'Location'

contains (selected) Enter or select a value...

OK Cancel

Order ID	Order Date	CustomerName	Location	Amount	Profit
1	B-26100				
2	B-26100				
3	B-26100				
4	B-26099				
5	B-26099				
6	B-26099				
7	B-26099				
8	B-26098				
9	B-26098				
10	B-26098				
11	B-26098				
12	B-26098				
13	B-26098				
14	B-26096				
15	B-26097				
16	B-26097	28-03-2019	Vini	Bangalore, Karnataka	19.00
17	B-26096	28-03-2019	Atharv	Kolkata, West Bengal	88.00
18	B-26097	28-03-2019	Vini	Bangalore, Karnataka	39.00
19	B-26096	28-03-2019	Atharv	Kolkata, West Bengal	264.00
20	B-26096	28-03-2019	Atharv	Kolkata, West Bengal	451.00
21	B-26097	28-03-2019	Vini	Bangalore, Karnataka	185.00
22	B-26097	28-03-2019	Vini	Bangalore, Karnataka	14.00
23	B-26097	28-03-2019	Vini	Bangalore, Karnataka	671.00

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Group By Use First Row as Headers Count Rows

Transpose Reverse Rows Detect Data Type Fill Rename Pivot Column Convert to List

Unpivot Columns Split Column Format Text Column Merge Columns Extract Parse

Statistics Standard Scientific Trigonometry Rounding Information

Date Time Duration Run R script Run Python script

Queries [4]

- List of Orders
- Order Details
- Sales target
- Orders Data

Table

fx = Table.SelectRows("#Sorted Rows", each ([Location] = "Chennai, Tamil Nadu"))

	Order ID	Order Date	CustomerName	Location	Amount	Profit
1	B-26081	22-03-2019	Aarushi	Chennai, Tamil Nadu	79.00	
2	B-26081	22-03-2019	Aarushi	Chennai, Tamil Nadu	169.00	
3	B-26081	22-03-2019	Aarushi	Chennai, Tamil Nadu	359.00	
4	B-26081	22-03-2019	Aarushi	Chennai, Tamil Nadu	93.00	
5	B-26081	22-03-2019	Aarushi	Chennai, Tamil Nadu	24.00	
6	B-26081	22-03-2019	Aarushi	Chennai, Tamil Nadu	637.00	
7	B-26018	14-02-2019	Aarushi	Chennai, Tamil Nadu	61.00	
8	B-26018	14-02-2019	Aarushi	Chennai, Tamil Nadu	326.00	
9	B-26008	09-02-2019	Kalyani	Chennai, Tamil Nadu	22.00	
10	B-26008	09-02-2019	Kalyani	Chennai, Tamil Nadu	206.00	
11	B-26008	09-02-2019	Kalyani	Chennai, Tamil Nadu	10.00	
12	B-26008	09-02-2019	Kalyani	Chennai, Tamil Nadu	57.00	
13	B-25860	15-11-2018	Akshay	Chennai, Tamil Nadu	112.00	
14	B-25788	21-09-2018	Dinesh	Chennai, Tamil Nadu	12.00	
15	B-25716	11-07-2018	Surabhi	Chennai, Tamil Nadu	58.00	
16	B-25698	23-06-2018	Amisha	Chennai, Tamil Nadu	207.00	
17	B-25698	23-06-2018	Amisha	Chennai, Tamil Nadu	27.00	
18	B-25698	23-06-2018	Amisha	Chennai, Tamil Nadu	65.00	
19	B-25698	23-06-2018	Amisha	Chennai, Tamil Nadu	87.00	
20	B-25698	23-06-2018	Amisha	Chennai, Tamil Nadu	7.00	
21	B-25698	23-06-2018	Amisha	Chennai, Tamil Nadu	516.00	
22	B-25608	08-04-2018	Aarushi	Chennai, Tamil Nadu	257.00	
23	B-25608	08-04-2018	Aarushi	Chennai, Tamil Nadu	856.00	

11 COLUMNS, 25 ROWS Column profiling based on entire data set

PREVIEW DOWNLOADED AT 10:23 PM

30°C Sunny

Search

ENG IN 10:23 PM 12-09-2026

Query Settings

PROPERTIES

Name

Orders Data

All Properties

APPLIED STEPS

- Source
- Expanded Order Details
- Sorted Rows
- Filtered Rows

Explanation: Orders Data was sorted by Order Date in descending order and filtered for Tamil Nadu using the Location field.

Question / Requirement

Duplicate the “Order Details” table and calculate the count of each Order ID, average profit by Category or total amount by Sub-Category.

The screenshot shows the Power Query Editor interface. The 'Group By' dialog box is open, allowing the user to specify the column to group by and the desired output. The 'Basic' tab is selected, and 'Order ID' is chosen as the column to group by. The 'Count' operation is selected, and the new column name is 'Count'. The background table shows data for various orders, including Order ID, Amount, Profit, Quantity, Category, and Sub-Category. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list, which includes 'Added Conditional Column'.

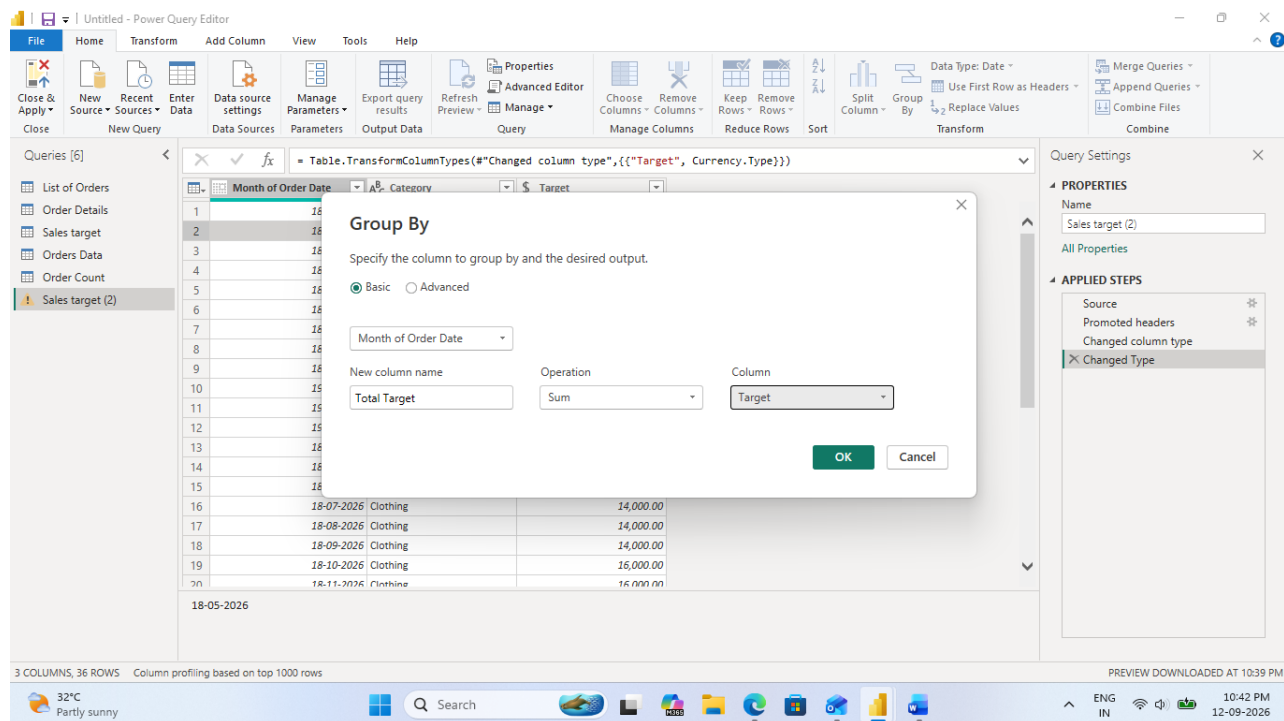
The screenshot shows the Power Query Editor interface after the 'Group By' operation. The table now has two columns: 'Order ID' and 'Order Count'. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list, which includes 'Grouped Rows'.

Order ID	Order Count
B-25601	4
B-25602	5
B-25603	8
B-25604	2
B-25605	1
B-25606	1
B-25607	1
B-25608	4
B-25609	2
B-25610	6
B-25611	1
B-25612	1
B-25613	1
B-25614	2
B-25615	1
B-25616	4
B-25617	1
B-25618	2
B-25619	1
B-25620	1

Explanation: Order Details was duplicated and grouped by Order ID using Count Rows, producing an Order Count for each Order ID. This satisfies the count-of-each-Order-ID option in the requirement.

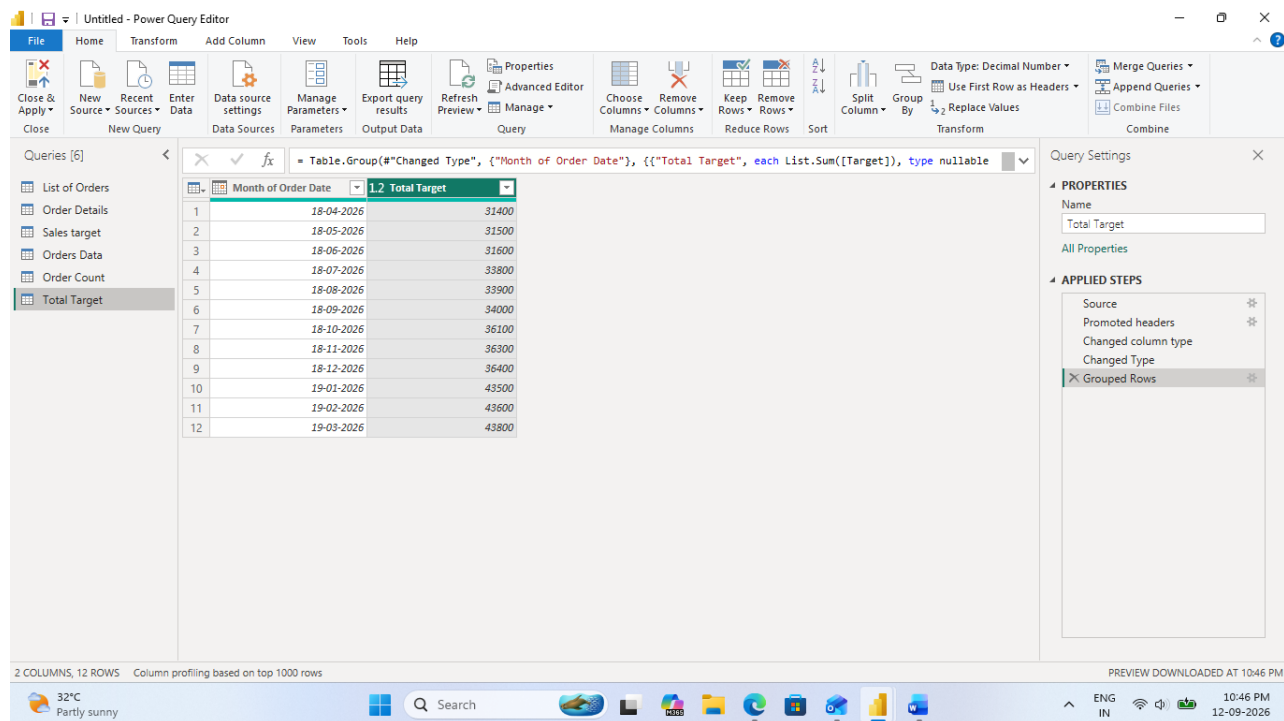
Question / Requirement

Duplicate the “Sales Target” table and aggregate the total target amount by Month of Order Date.



The screenshot shows the Power Query Editor interface. The main data table has columns: Month of Order Date, Category, and Target. A 'Group By' dialog box is open, showing the configuration for grouping by 'Month of Order Date' and calculating the 'Sum' of the 'Target' column. The new column name is 'Total Target'. The background table shows data for various months and categories, with target values.

Month of Order Date	Category	Target
18-04-2026		31400
18-05-2026		31500
18-06-2026		31600
18-07-2026		33800
18-08-2026		33900
18-09-2026		34000
18-10-2026		36100
18-11-2026		36300
18-12-2026		36400
19-01-2026		43500
19-02-2026		43600
19-03-2026		43800



The screenshot shows the final result of the grouping operation. The data is grouped by 'Month of Order Date', and the 'Total Target' column is calculated, showing monthly totals. The background table shows the grouped data with the 'Total Target' column.

Month of Order Date	1.2 Total Target
18-04-2026	31400
18-05-2026	31500
18-06-2026	31600
18-07-2026	33800
18-08-2026	33900
18-09-2026	34000
18-10-2026	36100
18-11-2026	36300
18-12-2026	36400
19-01-2026	43500
19-02-2026	43600
19-03-2026	43800

Explanation: Sales Target was duplicated and grouped by Month of Order Date using Sum of Target, producing monthly total target values.

Question / Requirement

Establish a relationship between the “List of Orders” and “Order Details” tables using the ‘Order ID’ column.

Edit relationship

Select tables and columns that are related.

From table: Order Details

Category	Order ID	Profit	Profit Margin	Profit Status	Quantity	Sub-Categ
Clothing	B-25603	1	0.08333333333...	Profit	2	Hankerc
Clothing	B-25608	23	0.0894941634...	Profit	5	Hankerc
Clothing	B-25615	20	0.2941176470...	Profit	5	Hankerc

To table: Sales target

Category	Month of Ord...
Furniture	18 April 2026
Furniture	18 May 2026
Furniture	18 June 2026

Cardinality: Many to many (*:*)

Cross-filter direction: Single ('Sales target' filters 'Order Details')

☒ Make this relationship active

☐ Assume referential integrity

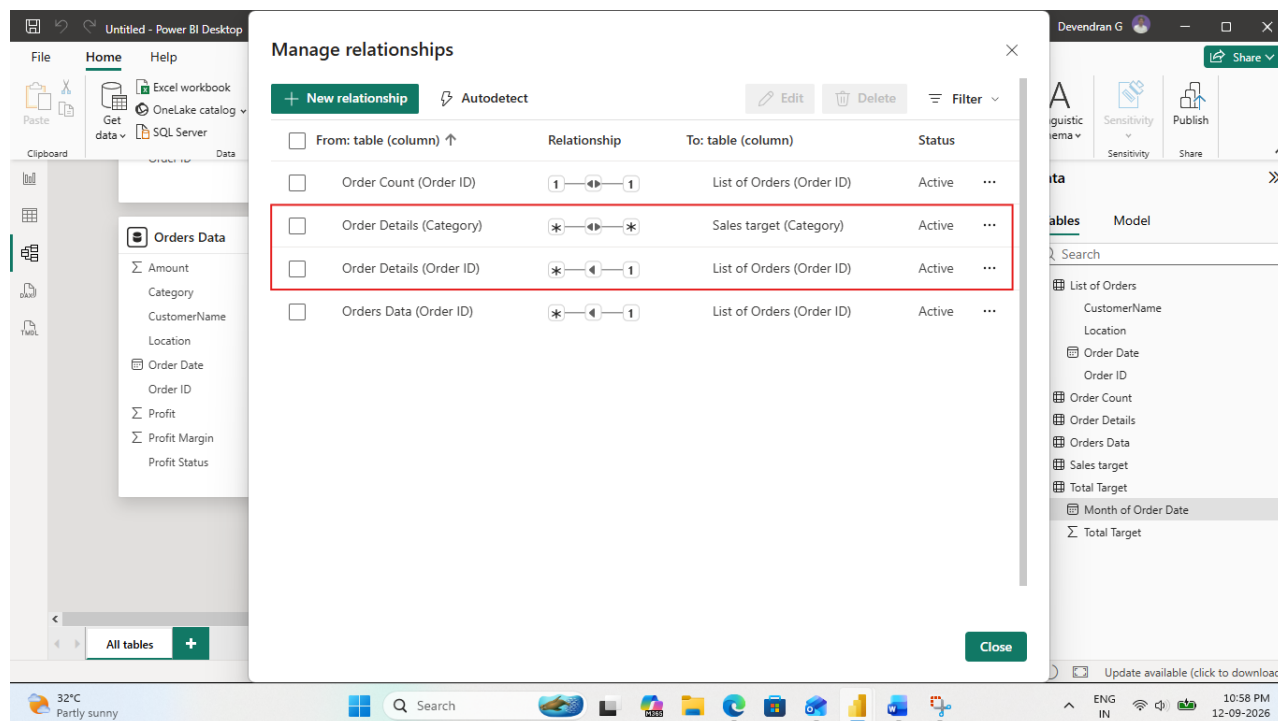
☐ Apply security filter in both directions

Save Cancel

Explanation: The Order ID relationship was created between List of Orders and Order Details. It is active and supports the one-to-many order-to-detail structure.

Question / Requirement

Build a relationship between the “Order Details” and “Sales Target” tables based on the ‘Category’ column. Click “Manage relationships” and ensure this relationship is active.



Explanation: The Category relationship between Order Details and Sales Target was made active. The relationship is configured as many-to-many because Category values repeat in both tables, with single-direction filtering from Sales Target to Order Details.

Final Submission Note

Main file: Submit the completed Power BI .pbix file as the primary assignment artifact. This document is supporting evidence showing each assignment point with the relevant screenshot and a concise explanation.

Before final submission, correct the Location column order to City, State.